
x
[B,n,H]
Wqkv
Wg
Q, K, V = W·x
+ shortconv
gates βt, λ̄t
= Wg·x
shift
φ(qt)
xt = φ(kt−1)
(RAW shift)
NLMS
NLMS
ηt =
βt/(‖xt‖²+λt+ε)
outer product
St = γtSt−1 +
ηt·xt·vtT
γt = 1 −
min(ηtλt, 1−ε)
read
next
carry
ot = StT·φ(qt)
(read-after-write)
St−1
[dx,dv]
Wo
O = Wo·ot
[B,n,H]